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Studierichting: Informatica
Oefeningenreeks 4A nr. 46.7

$$\int \frac{x^5 - x^4 + 4x + 4}{x^4 + 4} dx$$

We voeren eerst de euclidische deling uit:

$$\frac{x^5 - x^4 + 4x + 4}{x^4 + 4} = x - 1 + \frac{8}{x^4 + 4}$$

Dus:

$$\begin{aligned} \int \frac{x^5 - x^4 + 4x + 4}{x^4 + 4} dx &= \int (x - 1) dx + \int \frac{8}{x^4 + 4} dx \\ &= \frac{x^2}{2} - x + \int \frac{8}{x^4 + 4} dx \end{aligned}$$

We ontbinden de noemer:

$$\begin{aligned} x^4 + 4 &= x^4 + 4x^2 + 4 - 4x^2 \\ &= (x^2 + 2)^2 - (2x)^2 \\ &= (x^2 - 2x + 2)(x^2 + 2x + 2) \end{aligned}$$

Dus:

$$\int \frac{8}{x^4 + 4} dx = \int \frac{8}{(x^2 - 2x + 2)(x^2 + 2x + 2)} dx$$

Volgens de regel van Fuss schrijven we:

$$\frac{8}{(x^2 - 2x + 2)(x^2 + 2x + 2)} = \frac{Ax + B}{x^2 - 2x + 2} + \frac{Cx + D}{x^2 + 2x + 2}$$

We bepalen nu A, B, C, D met de regel van Fuss.

Voor $x^2 + 2x + 2$ nemen we een nulpunt:

$$x = -1 + i$$

Dan geldt:

$$\begin{aligned} C(-1 + i) + D &= \frac{8}{x^2 - 2x + 2} \Big|_{x=-1+i} \\ -C + Ci + D &= \frac{8}{(-1 + i)^2 - 2(-1 + i) + 2} \\ -C + Ci + D &= \frac{8}{4 - 4i} \end{aligned}$$

$$\begin{aligned}
-C + Ci + D &= \frac{2}{1-i} \\
-C + Ci + D &= \frac{2(1+i)}{(1-i)(1+i)} \\
-C + Ci + D &= 1+i
\end{aligned}$$

Hieruit volgt:

$$C = 1$$

$$-C + D = 1$$

$$D = 2$$

Voor $x^2 - 2x + 2$ nemen we een nulpunt:

$$x = 1 + i$$

Dan geldt:

$$\begin{aligned}
A(1+i) + B &= \frac{8}{x^2 + 2x + 2} \Big|_{x=1+i} \\
A + Ai + B &= \frac{8}{(1+i)^2 + 2(1+i) + 2} \\
A + Ai + B &= \frac{8}{4 + 4i} \\
A + Ai + B &= \frac{2}{1+i} \\
A + Ai + B &= \frac{2(1-i)}{(1+i)(1-i)} \\
A + Ai + B &= 1-i
\end{aligned}$$

Hieruit volgt:

$$A = -1$$

$$A + B = 1$$

$$B = 2$$

Dus:

$$\frac{8}{(x^2 - 2x + 2)(x^2 + 2x + 2)} = \frac{-x + 2}{x^2 - 2x + 2} + \frac{x + 2}{x^2 + 2x + 2}$$

Bijgevolg:

$$\int \frac{x^5 - x^4 + 4x + 4}{x^4 + 4} dx = \frac{x^2}{2} - x + \int \frac{-x + 2}{x^2 - 2x + 2} dx + \int \frac{x + 2}{x^2 + 2x + 2} dx$$

We werken de eerste integraal uit:

$$\begin{aligned} I_1 &= \int \frac{-x + 2}{x^2 - 2x + 2} dx \\ &= -\frac{1}{2} \int \frac{2x - 4}{x^2 - 2x + 2} dx \\ &= -\frac{1}{2} \int \frac{2x - 2 - 2}{x^2 - 2x + 2} dx \\ &= -\frac{1}{2} \int \frac{2x - 2}{x^2 - 2x + 2} dx + \int \frac{1}{x^2 - 2x + 2} dx \\ &= -\frac{1}{2} \ln(x^2 - 2x + 2) + \int \frac{1}{(x - 1)^2 + 1} dx \\ &= -\frac{1}{2} \ln(x^2 - 2x + 2) + \text{Bgtan}(x - 1) \end{aligned}$$

We werken de tweede integraal uit:

$$\begin{aligned} I_2 &= \int \frac{x + 2}{x^2 + 2x + 2} dx \\ &= \frac{1}{2} \int \frac{2x + 4}{x^2 + 2x + 2} dx \\ &= \frac{1}{2} \int \frac{2x + 2 + 2}{x^2 + 2x + 2} dx \\ &= \frac{1}{2} \int \frac{2x + 2}{x^2 + 2x + 2} dx + \int \frac{1}{x^2 + 2x + 2} dx \\ &= \frac{1}{2} \ln(x^2 + 2x + 2) + \int \frac{1}{(x + 1)^2 + 1} dx \\ &= \frac{1}{2} \ln(x^2 + 2x + 2) + \text{Bgtan}(x + 1) \end{aligned}$$

Dus de totale primitieve is:

$$\begin{aligned} &\int \frac{x^5 - x^4 + 4x + 4}{x^4 + 4} dx \\ &= \frac{x^2}{2} - x - \frac{1}{2} \ln(x^2 - 2x + 2) + \text{Bgtan}(x - 1) \\ &\quad + \frac{1}{2} \ln(x^2 + 2x + 2) + \text{Bgtan}(x + 1) + C \end{aligned}$$

References